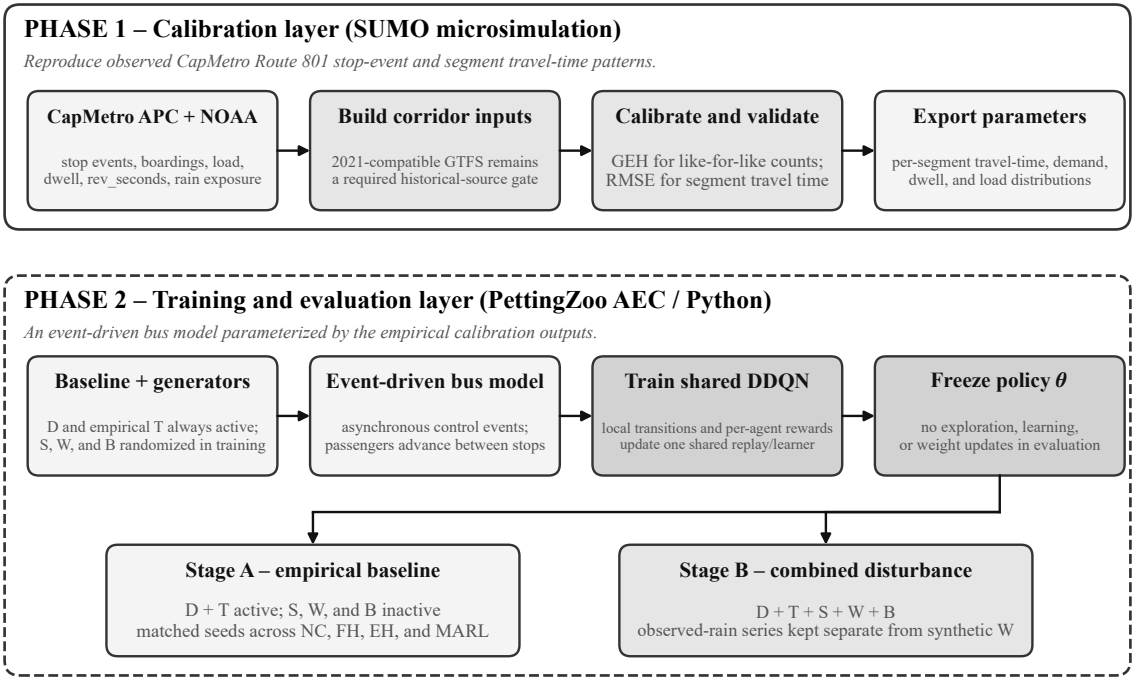

PHASE 1 – Calibration layer (SUMO microsimulation)
Reproduce observed CapMetro Route 801 stop-event and segment travel-time patterns.
CapMetro APC + NOAA
stop events, boardings, load, dwell, rev_seconds, rain exposure
Build corridor inputs
2021-compatible GTFS remains a required historical-source gate
Calibrate and validate
GEH for like-for-like counts; RMSE for segment travel time
Export parameters
per-segment travel-time, demand, dwell, and load distributions
PHASE 2 – Training and evaluation layer (PettingZoo AEC / Python)
An event-driven bus model parameterized by the empirical calibration outputs.
Baseline + generators
D and empirical T always active; S, W, and B randomized in training
Event-driven bus model
asynchronous control events; passengers advance between stops
Train shared DDQN
local transitions and per-agent rewards update one shared replay/learner
Freeze policy θ
no exploration, learning, or weight updates in evaluation
Stage A – empirical baseline
D + T active; S, W, and B inactive
matched seeds across NC, FH, EH, and MARL
Stage B – combined disturbance
D + T + S + W + B
observed-rain series kept separate from synthetic W